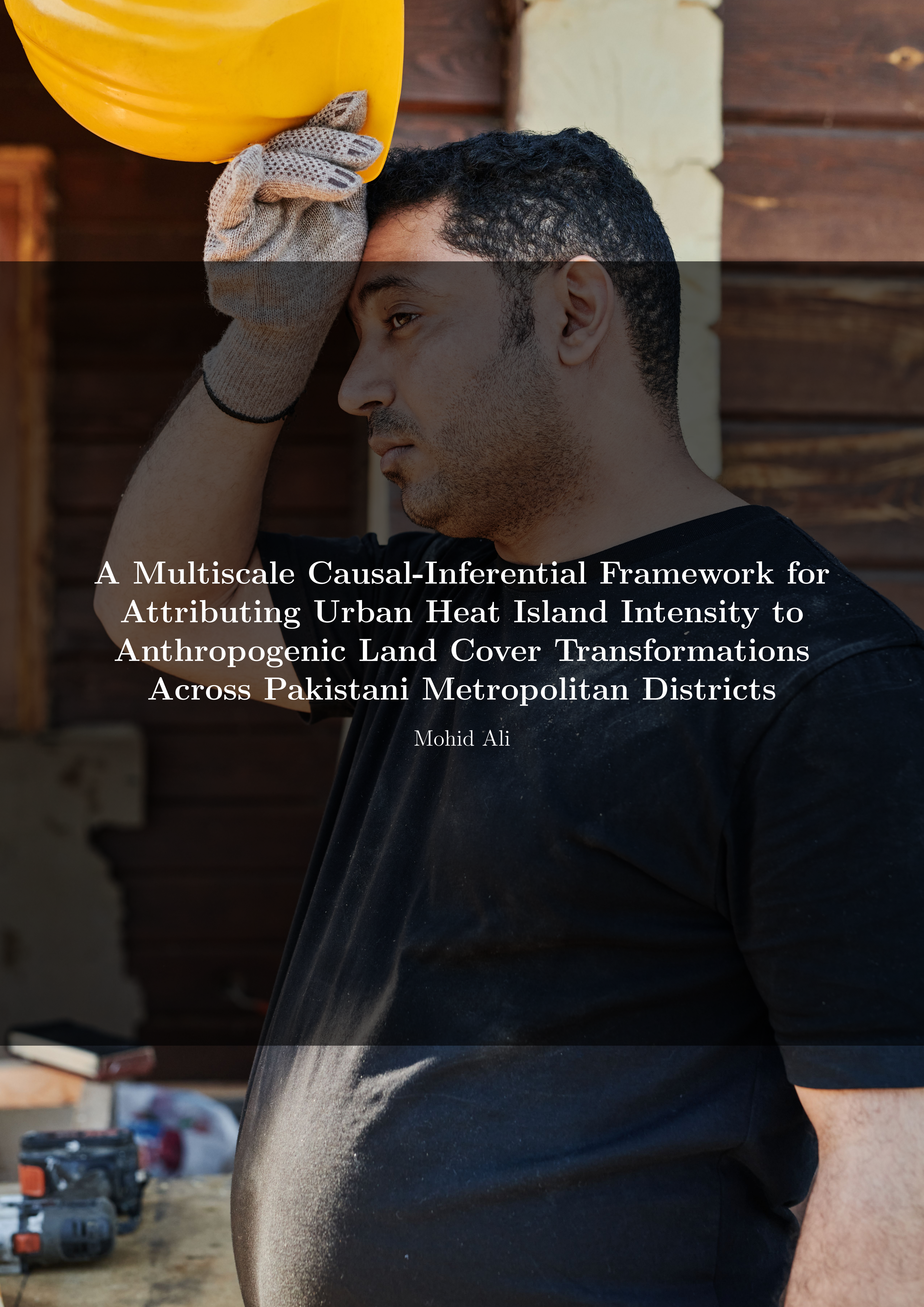

A Multiscale Causal-Inferential Framework for Attributing Urban Heat Island Intensity to Anthropogenic Land Cover Transformations Across Pakistani Metropolitan Districts

Mohid Ali

Abstract

Urban Heat Islands (UHIs) intensify heat exposure in cities and are a growing public health and climate concern, yet Pakistan's cities remain almost entirely absent from the UHI literature. This project builds a multi-city UHI analysis framework covering five major Pakistani cities – Karachi, Lahore, Islamabad, Peshawar, and Faisalabad – combining satellite remote sensing, machine learning, and causal inference. A ten-year, district-level dataset is assembled from Google Earth Engine (Landsat, Sentinel-2, MODIS), Open-Meteo, WorldPop, Copernicus Global Land Cover, OpenStreetMap, and national poverty statistics. Three predictive models are trained: a Random Forest for driver analysis, a U-Net convolutional network for pixel-level heat mapping, and an LSTM for short-term forecasting. A causal directed acyclic graph is estimated using ordinary least squares, propensity score matching, and G-computation to move beyond correlation and quantify the cooling effect of vegetation, with particular attention to whether that effect differs by income level and season. The trained U-Net is then repurposed as a physics-informed simulator to evaluate two concrete interventions – green cover expansion and cool roofs – and the entire framework is exposed through an interactive Streamlit dashboard. Results show that vegetation has a robust, causally supported cooling effect that is roughly three times larger in low-income districts and four times larger in summer than in winter, and that cool roofs outperform green cover on cooling magnitude in every city studied.

Contents

1 Introduction

Urban Heat Islands occur when built-up areas absorb and retain more heat than their surrounding rural areas, raising local temperatures and compounding the effects of climate change. In rapidly urbanizing, climate-vulnerable countries such as Pakistan, this effect can push already extreme summer temperatures into dangerous territory, with the greatest burden typically falling on low-income residents who have the least capacity to adapt. Despite this, Pakistani cities are severely underrepresented in the global UHI literature, and almost no existing work moves beyond simple correlation to ask whether interventions such as urban greening actually *cause* cooling, and for whom.

This project addresses that gap with a data-driven framework built around three questions: (1) where and when is urban heat most intense across Pakistan's major cities, (2) what causally drives it, and (3) what happens if a specific policy – planting more greenery or repainting rooftops – is applied. The framework combines supervised machine learning for prediction, causal inference for explanation, and a trained neural network repurposed as a simulator for policy evaluation, all delivered through an interactive dashboard.

2 Study Area and Data

2.1 Study Area

Five of Pakistan's largest and most climatically diverse cities were selected to capture variation in geography, elevation, and development pattern: the coastal megacity of Karachi, the flat agricultural plain of Lahore and Faisalabad, and the elevated cities of Islamabad and Peshawar near the northern hills.

City	Latitude	Longitude
Karachi	24.86	67.01
Lahore	31.55	74.35
Islamabad	33.72	73.06
Peshawar	34.01	71.57
Faisalabad	31.41	73.07

Table 1: Study cities and their approximate coordinates.

2.2 Data Sources

Data was assembled from seven public sources, each contributing a different layer of the analysis:

- **Google Earth Engine**: Landsat 8/9 Collection 2 Surface Reflectance (thermal band converted to Land Surface Temperature, NDVI, NDBI, albedo), Sentinel-2 10m NDVI/NDBI, and MODIS MOD11A1 as a daily LST supplement. Cloud-masked monthly median composites were built for 2015–2024.
- **Open-Meteo**: hourly air temperature, humidity, precipitation, and wind speed, aggregated to monthly city-level weather variables.
- **WorldPop**: 100m-resolution population density rasters.

- **Copernicus Global Land Cover and GHSL**: built-up, vegetation, and water classifications used to derive green space percentage and urban structure.
- **OpenStreetMap (via OSMnx)**: distances to green spaces, hospitals, and water bodies.
- **PBS / World Bank Pakistan Poverty Assessment**: district-level poverty headcount ratios, used both as a socioeconomic control and as the moderating variable in the heterogeneous treatment effect analysis.

2.3 Master Dataset Schema

All sources were merged into a single master table at the (city, district, month, year) level, spanning 2015–2024 across 30 districts – 3,600 rows in total, with no missing values after cleaning. Key columns include:

- **lst_mean / lst_max / lst_min**: Land Surface Temperature from Landsat (°C)
- **uhi_intensity**: district LST minus the city-month mean LST, i.e. how much hotter a district is than its own city's average that month
- **ndvi_mean**: vegetation index (−1 to +1, higher is greener)
- **ndbi_mean**: built-up index (higher indicates more concrete and impervious surface)
- **albedo_mean**: surface reflectivity (lower absorbs more heat)
- **impervious_fraction**: share of district area covered by roads and buildings
- **population_density**: from WorldPop
- **poverty_index**: from the PBS/World Bank poverty assessment
- **green_space_pct**: share of district area as parks or vegetation
- **dist_to_water**: distance to the nearest water body (km)
- **lst_trend**: long-run warming slope (°C per year, 2015–2024)
- **air_temp_mean, humidity_mean**: monthly weather aggregates from Open-Meteo

2.4 Data Preprocessing

Cloud-affected satellite readings (identified via the Landsat QA_PIXEL band) were linearly interpolated within each district's time series instead of being dropped, preserving the full monthly cadence. Satellite, weather, demographic, and socioeconomic tables were spatially and temporally joined on city, district, year, and month. All processed data passed a validation pass checking for missing values, physically plausible ranges, and expected correlations (for example, NDVI and NDBI were confirmed to be strongly negatively correlated, as vegetation and built-up land are largely mutually exclusive).

3 Methodology

3.1 Exploratory and Geospatial Analysis

Before modeling, the dataset was examined through univariate, temporal, spatial, and correlation-based exploratory analysis. Seasonal decomposition confirmed a strong annual heat cycle in every city, and spatial autocorrelation (Moran's I) was used to test whether high-UHI districts cluster geographically instead of appearing at random. A composite vulnerability index – combining UHI intensity, poverty, population density, and distance to water – was constructed to identify districts where heat exposure and socioeconomic vulnerability overlap, and these overlap districts were flagged as *hotspots* for use in later phases.

3.2 Predictive Modeling

Three complementary models were trained, each suited to a different question:

- **Random Forest (driver analysis)**: trained on 22 tabular features (17 physical/socioeconomic variables plus city indicators, excluding `lst_mean` itself to avoid leakage) to identify which factors are most predictive of UHI intensity, using SHAP values for interpretability.
- **U-Net convolutional network (spatial heat mapping)**: a compact two-level U-Net that takes a 64×64 pixel tile of NDVI, NDBI, and albedo as input and predicts the corresponding Land Surface Temperature tile, learning the pixel-level relationship between land cover and heat.
- **LSTM (temporal forecasting)**: a two-layer recurrent network that consumes the past 24 months of LST and climate variables for a district and forecasts next-month LST, used to project near-term heat trajectories.

3.3 Causal Inference

This is the central contribution of the project: most UHI research, in Pakistan and globally, reports correlations between vegetation, built-up land, and heat without establishing causation. A structural causal model was defined as a directed acyclic graph (DAG) with nine nodes – city and air temperature as confounders, poverty and impervious fraction as socioeconomic and structural drivers, green space and NDVI as vegetation mediators, NDBI and albedo as urban structure mediators, and UHI intensity as the outcome.

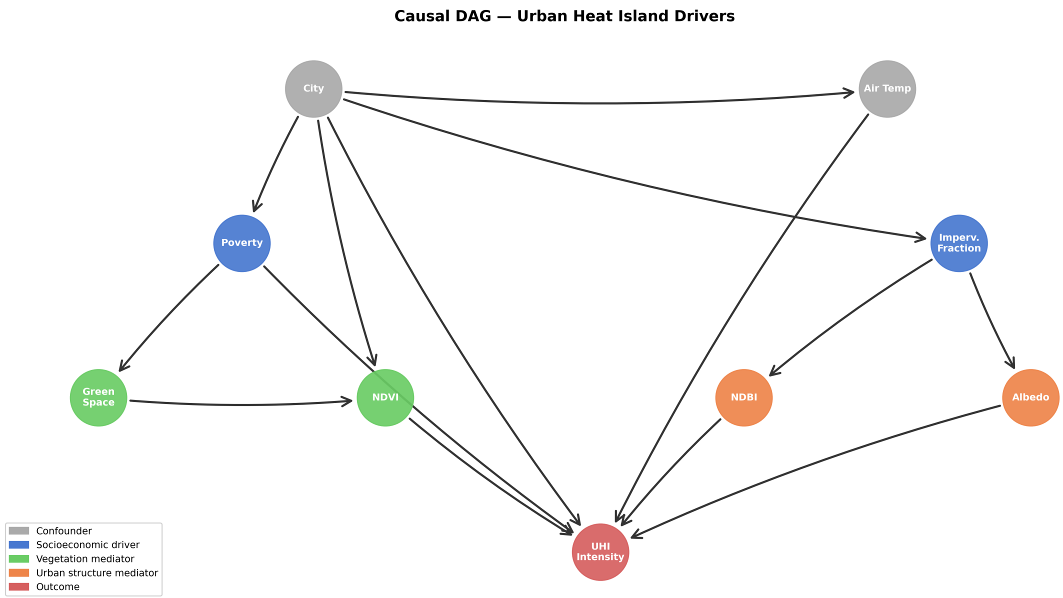

Figure 1: Causal directed acyclic graph underlying the treatment effect estimates, showing confounders (gray), socioeconomic drivers (blue), vegetation and urban structure mediators (green, orange), and the outcome (red).

Using this graph, three causal treatments (NDVI, impervious fraction, and poverty) were estimated with backdoor-adjusted ordinary least squares and propensity score matching, controlling for city and air temperature. Every estimate was stress-tested with placebo permutation and random common cause refutation tests, which the vegetation and poverty effects passed.

To move from average effects to actionable, district-specific answers, G-computation with the trained Random Forest was used to simulate four realistic policy scenarios (for example, raising every hotspot district's NDVI to its city's average), and a set of heterogeneous treatment effect models tested whether the cooling effect of vegetation differs systematically by district poverty level, city, and season.

3.4 Simulation Engine

The trained U-Net is repurposed as a lightweight, physics-informed simulator: for a given city, the input tile's NDVI channel is increased (green cover) or its albedo channel is increased (cool roofs), and the model is re-run to predict the resulting LST. The difference between the baseline and post-intervention prediction gives an estimated cooling effect in degrees Celsius, applied specifically to each city's hotspot districts (those with above-median LST).

4 Evaluation Metrics

All predictive models used a strictly chronological train/validation/test split (2015–2021 / 2022 / 2023–2024), since shuffling time-series data would let the model see the future during training and produce misleadingly optimistic results. The Random Forest and U-Net were evaluated with R^2 and Mean Absolute Error (MAE) on held-out test data, and hyperparameters for the Random Forest were tuned with 50 trials of Optuna optimization.

The LSTM was evaluated with per-city MAE, since forecast difficulty varies meaningfully by city.

Causal estimates cannot be evaluated with a train/test split, since there is no ground truth counterfactual to check against. Instead, confidence in each causal claim rests on refutation testing: a placebo permutation test (does the effect disappear when the treatment is randomly shuffled?) and a random common cause test (does the effect survive the addition of an irrelevant confounder?). Both tests were run for every reported causal effect, and effects that did not survive refutation – most notably the raw impervious-fraction-to-UHI relationship, which is confounded by elevation in Islamabad and Peshawar – are explicitly flagged as unreliable.

5 Results Summary

5.1 Predictive Model Performance

The Random Forest driver model reached a test R^2 of 0.69 with a mean absolute error of under 1°C, confirming that the available physical and socioeconomic features explain most of the variation in UHI intensity. The U-Net spatial heat mapper achieved a comparable pixel-level test MAE of roughly 1.3°C, validating it as a reasonable basis for the simulation engine described below. The LSTM forecaster's accuracy varied by city, performing best in Islamabad and worst in coastal Karachi, where maritime weather introduces additional month-to-month variability that is harder to forecast from land-surface signals alone.

5.2 Causal Findings

The causal analysis confirms that vegetation has a real, causally supported cooling effect on Land Surface Temperature, and this effect survived both refutation tests. More importantly, poorer districts benefit substantially more from a given increase in vegetation than wealthier ones, and the cooling effect is roughly four times stronger in summer than in winter – meaning greening delivers the most benefit exactly when and where it is needed most. The impervious-fraction-to-heat relationship, by contrast, was found to be confounded by elevation differences between cities and is reported as a limitation.

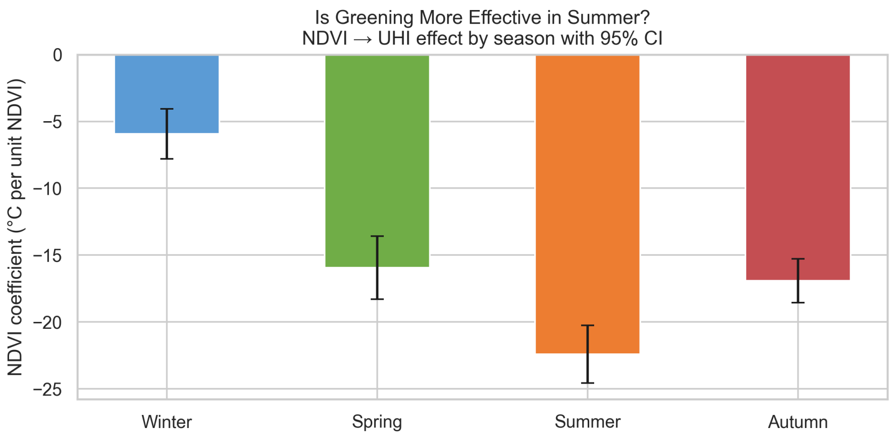

Figure 2: Estimated cooling effect of a 1-unit NDVI increase on UHI intensity, by season. The effect is roughly four times stronger in summer than in winter.

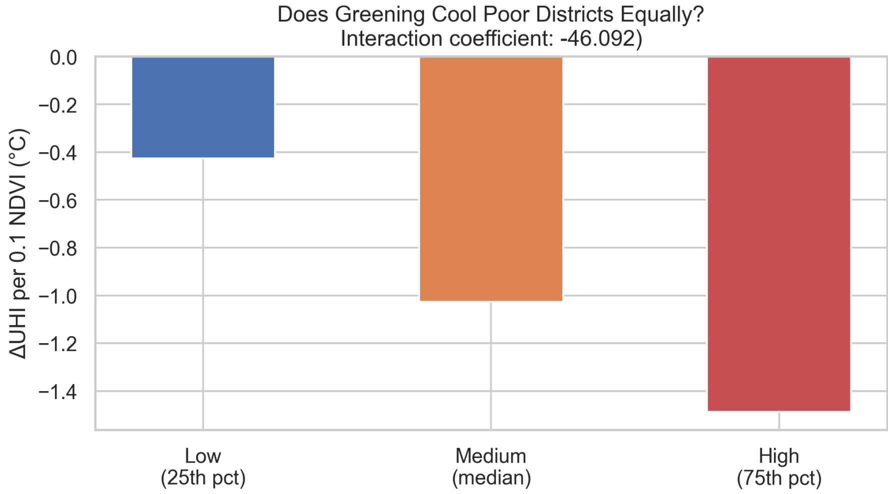

Figure 3: Estimated cooling effect of a 0.1-unit NDVI increase on UHI intensity, by district poverty level. Poorer districts show a substantially larger cooling response.

5.3 Simulated Interventions

Simulating both interventions across all five cities' hotspot districts shows that cool roofs produce a larger temperature reduction than green cover in every city tested, though green cover has a lower estimated cost and can be targeted more precisely at low-income neighborhoods. Lahore's cool-roof scenario was estimated to benefit the largest population of any city-intervention pair, largely because Lahore's hotspot districts are more densely populated than those identified in the other four cities.

Ranked Intervention Table — Green Cover vs Cool Roofs

City	Intervention	Cooling (°C)	Population Benefited	Cost Tier	Complexity
Karachi	Cool Roofs	10.65	307,752	Med	Low
Karachi	Green Cover	5.86	307,752	Low	Med
Lahore	Cool Roofs	5.41	8,177,076	Med	Low
Lahore	Green Cover	3.72	8,177,076	Low	Med
Islamabad	Cool Roofs	6.25	1,594,045	Med	Low
Islamabad	Green Cover	4.72	1,594,045	Low	Med
Peshawar	Cool Roofs	5.96	742,299	Med	Low
Peshawar	Green Cover	3.68	742,299	Low	Med
Faisalabad	Cool Roofs	6.11	3,629,302	Med	Low
Faisalabad	Green Cover	3.63	3,629,302	Low	Med

Figure 4: Ranked comparison of the two simulated interventions across all five cities, showing predicted cooling, population benefited, cost tier, and implementation complexity.

6 Model Interpretability

Two complementary interpretability strategies were used, matched to each model's purpose. For the Random Forest, SHAP (SHapley Additive exPlanations) values quantify how much each feature pushes a given prediction up or down, and consistently ranked vegetation, impervious fraction, and poverty among the most influential drivers of UHI intensity – aligning with prior domain expectations. For the causal component, interpretability is built in by design: the DAG is an explicit, human-readable structural model, so every causal claim in this project can be traced back to a specific, inspectable assumption about how the variables relate. The U-Net and LSTM, by contrast, are used purely for prediction and simulation, with their outputs validated against held-out data instead of interpreted feature-by-feature.

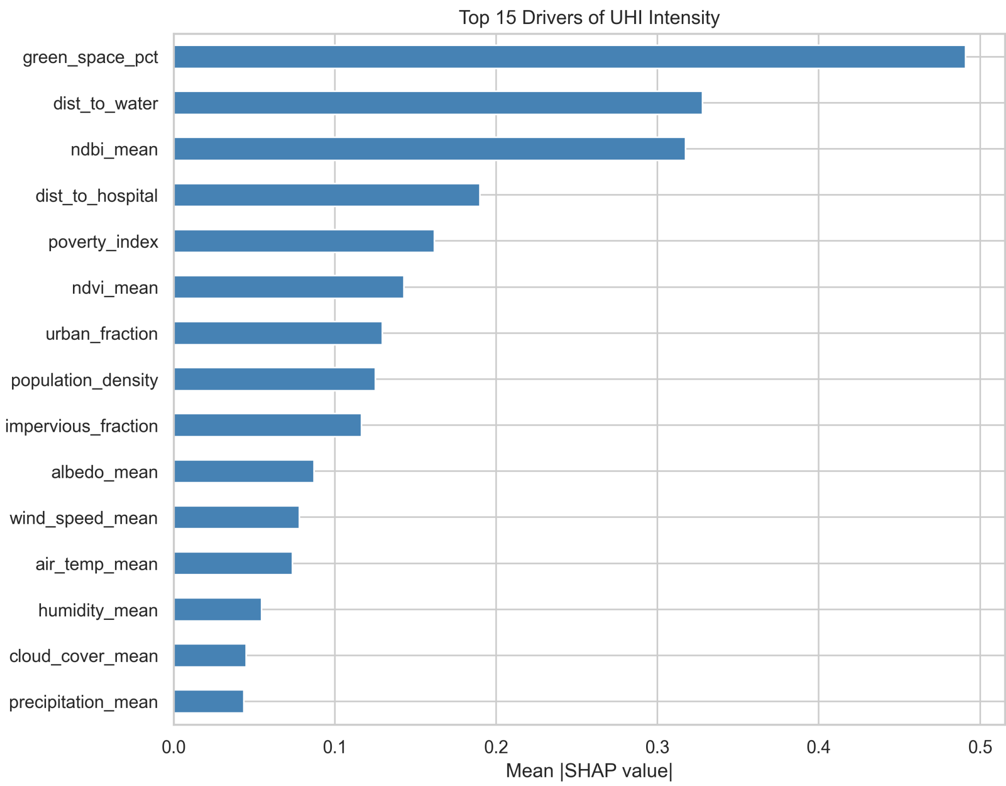

Figure 5: SHAP feature importance for the Random Forest driver model, ranking the strongest predictors of UHI intensity.

7 Streamlit Dashboard

All of the above – the raw data, the three predictive models, the causal findings, and the simulation engine – are exposed through a live, interactive dashboard built with Streamlit, so that the analysis can be explored. The dashboard is deployed at `https://uhi-pakistan.streamlit.app` and is publicly accessible. It is organized into five pages, described below.

7.1 City Overview

The landing page reports city-level key metrics – peak district UHI intensity, peak Land Surface Temperature, mean NDVI, and hotspot district count – alongside a live choropleth map that lets the user switch between UHI intensity, LST, NDVI, and other metrics at the district level.

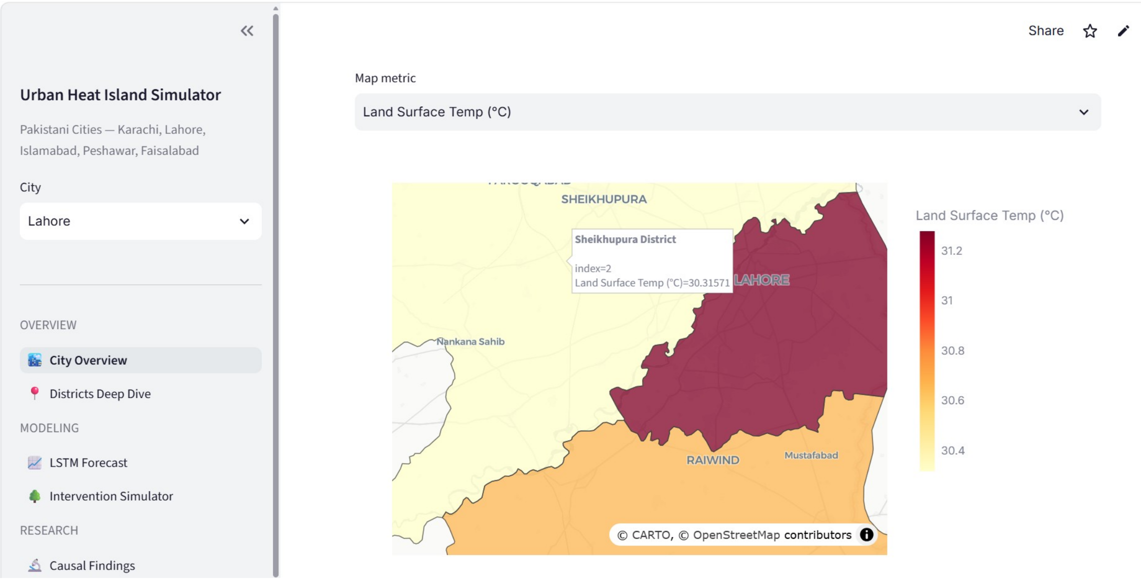

Figure 6: City Overview page, showing the district-level Land Surface Temperature choropleth for Lahore.

7.2 Districts Deep Dive

Selecting an individual district reveals its full ten-year temporal trend, a location map, an NDVI-versus-LST scatter plot with seasonal trendlines, and a breakdown of its composite vulnerability score by component.

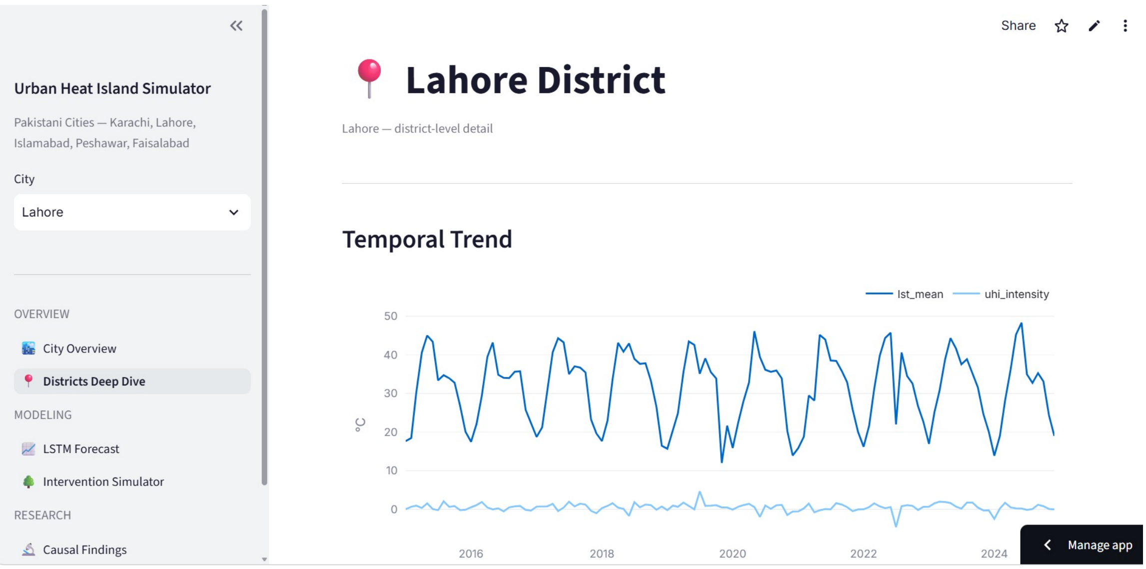

Figure 7: Districts Deep Dive page, showing the ten-year Land Surface Temperature and UHI intensity trend for a selected district.

7.3 LSTM Forecast

This page runs the trained LSTM forward autoregressively for a user-selected forecast horizon, displaying the projected LST trajectory alongside a heuristic uncertainty band that widens with each forecast step.

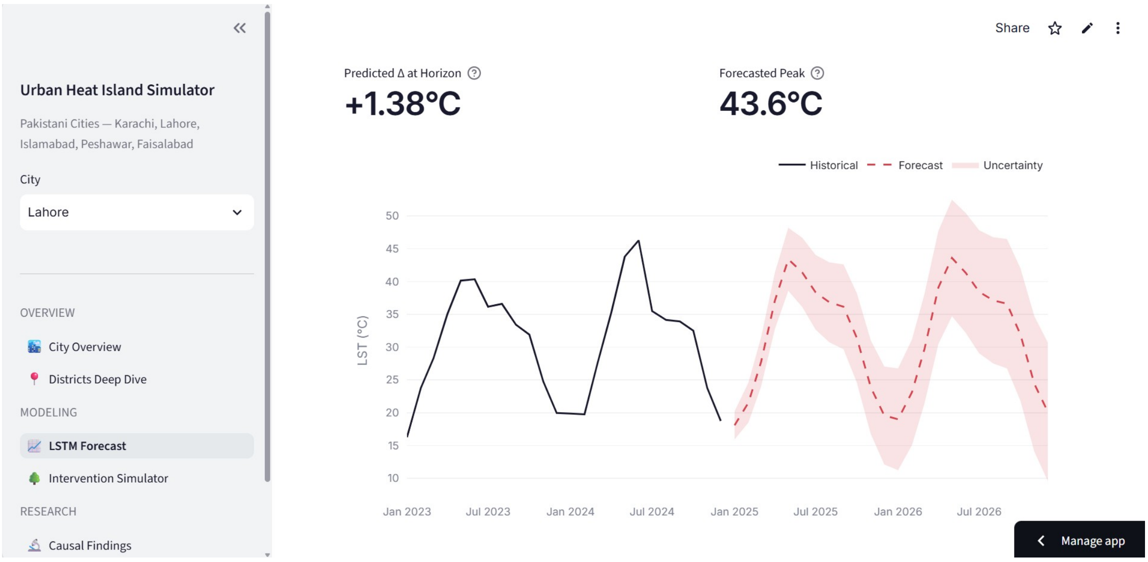

Figure 8: LSTM Forecast page, showing a multi-year Land Surface Temperature projection with historical data, forecast, and uncertainty band.

7.4 Intervention Simulator

The simulator lets a user choose a city, an intervention (green cover or cool roofs), and a coverage level, then runs live U-Net inference on that city's hotspot tiles and reports predicted cooling, population benefited, cost tier, and implementation complexity, alongside a before/after/delta heat map.

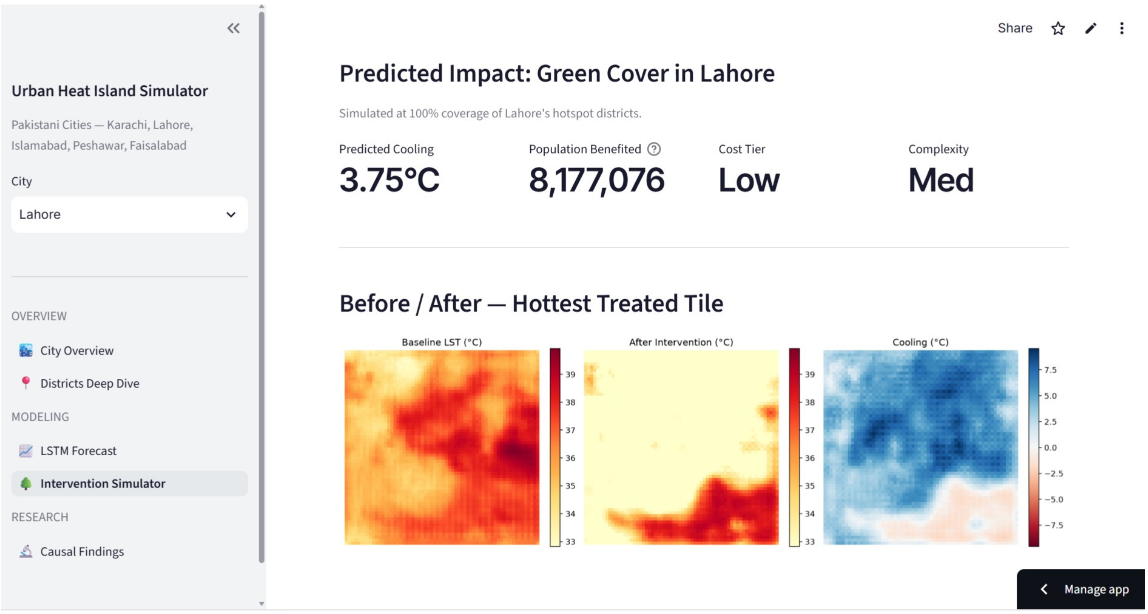

Figure 9: Intervention Simulator page, showing predicted impact and before/after Land Surface Temperature maps for a green cover intervention in Lahore.

7.5 Causal Findings

The final page presents the causal DAG, the estimated average treatment effects, the counterfactual scenario results, and the heterogeneous treatment effect breakdowns by poverty, city, and season, all as live, explorable charts.

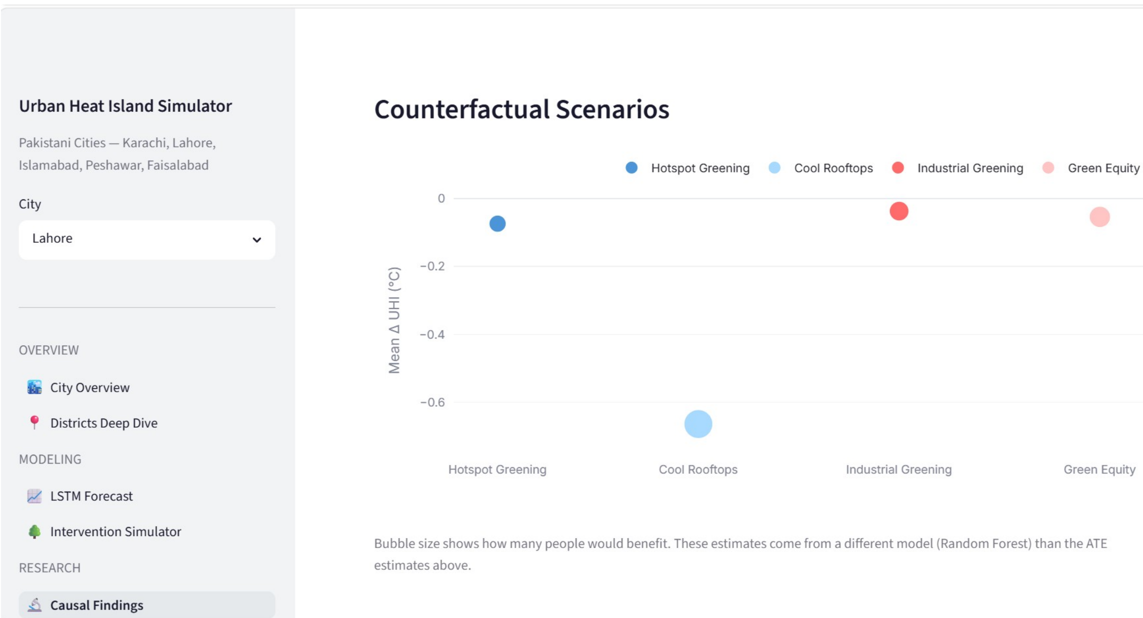

Figure 10: Causal Findings page, showing the counterfactual scenario results for the four simulated interventions.

7.6 Accompanying Article

Alongside this technical report, the project's findings are presented in a long-form article for a general audience, published on Medium as *"What 10 Years of Satellite Data Reveal About Urban Heat in Pakistan's Cities"*. The article goes in depth on the project's key findings – the causal analysis of vegetation's cooling effect, the equity finding that greening cools poor neighborhoods roughly three times more than wealthy ones, and the simulated cooling from the green cover and cool roof interventions – explained in accessible, non-technical language for a general audience, and links directly to the live dashboard for readers who want to explore the results themselves.

8 Ethics and Limitations

This project is a research and educational tool, and its outputs require further engineering and feasibility study before they can justify a specific policy decision. Several specific limitations are worth highlighting.

First, UHI intensity in Islamabad and Peshawar is affected by elevation: their surrounding rural buffer zones include high-altitude terrain that is naturally cooler for reasons unrelated to urban development, which confounds any raw comparison between built-up land and heat in those two cities. This is why the impervious-fraction causal estimate was ultimately not reported as a reliable finding.

Second, all causal estimates in this project are derived from observational data; no randomized controlled experiment was possible at this scale. While the refutation tests (placebo permutation and random common cause) increase confidence that the vegetation and poverty effects are not spurious, they cannot rule out every possible unmeasured confounder, and the results should be read as strong supporting evidence, stopping short of definitive proof.

Third, the simulation engine assumes that the U-Net's learned relationship between land cover and temperature continues to hold for interventions larger than anything seen in the training data; large-scale greening or roofing programs should be validated with a pilot before being scaled city-wide. Poverty and demographic figures are also drawn from a national assessment with a fixed reference year and may not reflect current, more granular conditions. Finally, satellite-derived measurements carry inherent resolution and cloud-gap limitations, which were mitigated through interpolation but not eliminated.

9 Conclusion

This project delivers the first multi-city causal analysis of Urban Heat Islands in Pakistan, combining machine learning, causal inference, and an interactive simulation dashboard in a single framework. The central finding – that vegetation's cooling effect is both causally supported and substantially larger for low-income and summer conditions – offers a concrete, equity-aligned argument for prioritizing urban greening in heat mitigation policy, while the simulator makes it possible to compare that policy against cool roofs on cooling magnitude, population benefited, and cost. Future work includes expanding the dataset to more cities and a longer time horizon, and incorporating additional causal treatments.

References

[1] Google Earth Engine, *Landsat, Sentinel-2, and MODIS Data Catalog*. Available at: https://earthengine.google.com

[2] Open-Meteo, *Historical Weather API*. Available at: https://open-meteo.com

[3] WorldPop, *Global High-Resolution Population Density Rasters*. Available at: https://www.worldpop.org

[4] Copernicus Global Land Service, *Global Land Cover Layers*. Available at: https://land.copernicus.eu/global/products/lc

[5] Boeing, G. (2017). OSMnx: New Methods for Acquiring, Constructing, Analyzing, and Visualizing Complex Street Networks. *Computers, Environment and Urban Systems*.

[6] Pakistan Bureau of Statistics / World Bank, *Pakistan Poverty Assessment*, 2016.

[7] Sharma, A., & Kiciman, E. (2020). DoWhy: An End-to-End Library for Causal Inference. arXiv:2011.04216.

[8] Microsoft Research, *EconML: A Python Package for ML-Based Heterogeneous Treatment Effects Estimation*. Available at: https://github.com/py-why/EconML